The Origins of the STEM Gender Gap: Evidence from an Analysis of Secondary School Subject Selection

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Abstract

Secondary school subject choices are a key gateway to STEM (Science, Technology, Engineering and Maths), yet their determinants remain underexplored. Using the Growing Up in Ireland study, which followed participants from the age of 9 to the end of secondary education, we apply regression and decomposition analysis to trace how gender gaps in STEM participation develop. Girls are more likely than boys to take traditional academic STEM subjects such as science and maths; the large gaps instead lie in practical, technology-oriented subjects such as engineering and construction, which boys are 30 percentage points more likely to take and which strongly predict college STEM enrolment. Lower-secondary subject choices, made around age 13, account for over half of this upper-secondary gap. The lower-secondary gap is even larger, at 59 percentage points, yet rich age-9 measures of ability, comparative advantage, preferences, aspirations and family background explain only 3% of it. Nor do schools explain it: the gap is larger in mixed-sex than single-sex schools and persists within schools once subject availability and school sorting are absorbed by fixed effects. Machine learning methods with a much larger covariate set yield the same conclusions. The evidence points away from skill- and school-based explanations and towards mechanisms such as gender norms, identifying ages 9 to 13 as a critical window for policy intervention.

Keywords: STEM, Subject Choice, Gender, Secondary School

JEL Classification: I21 , I24

Acknowledgements. We are grateful to David Madden, Christopher Jepsen, Benjamin Elsner, Aedín Doris, Prachi Srivastava, and particularly Paul Devereux, as well as seminar participants at University College Dublin for helpful comments on earlier drafts of this paper.

1 Introduction

STEM fields play a critical role in driving economic growth, innovation, and productivity, contributing substantially to national competitiveness. Substantial gender disparities in STEM participation in higher education are well established and persist. These disparities have important implications for gender equity,

wage equality, and diversity in innovation-driven sectors, making them an important focus for policymakers.[1] As a result, a large literature has examined the determinants of gender gaps in STEM participation in higher education. A consistent finding across this literature is that secondary school subject choices strongly predict later STEM pathways (Card and Payne 2021; Chevalier 2011; Conger et al. 2021; Delaney and Devereux 2019; Gottfried and Bozick 2016; Granato 2023; McNally 2020; Tchuente 2016). This points to secondary school as a critical stage at which gender differences in STEM pathways emerge, and therefore a key setting for understanding the origins of these gaps.

Treating the "STEM gender gap" as a single phenomenon risks obscuring important differences across subjects. We find that in secondary school, female students are more likely than male students to take traditional academic STEM subjects such as biology and chemistry. In contrast, large gender gaps in favour of male students arise primarily in practical, technology-oriented subjects. This suggests that the relevant question is not simply why girls avoid STEM, but which types of STEM fields exhibit gender gaps and at what stage those gaps emerge. We address these questions descriptively in this paper.

Existing studies of secondary school subject choice have highlighted several possible explanations for gender differences, including psychological factors and preferences, such as self-confidence and appreciation for STEM subjects (Stokking 2000), and broader psychological and social influences (Mackenzie et al. 2021). School-related attributes and teaching practices also influence enrolment decisions (Smith and Hannan 2006). Schools may matter through peer effects: female students are more likely to select STEM subjects when exposed to high-achieving peers of the same sex (Mouganie and Wang 2020), whereas exposure to a larger share of girls in earlier grades can reduce the likelihood of choosing traditionally female-dominated subjects (Schneeweis and Zweimüller 2012). Finally, some research has emphasised the role of skills, which may shape gender gaps in two ways. Gaps may reflect differences in ability, although prior maths achievement alone does not fully explain them (Friedman-Sokuler and Justman 2016; Speer 2017). Alternatively, they may reflect sex-specific comparative advantage, whereby students specialise in the subjects in which they are relatively strongest, so that girls' relative strength in verbal skills may draw them away from STEM even when their maths achievement is high (Delaney and Devereux 2019; Goulas et al. 2024). Most existing literature focuses on the later stages of secondary school, and typically examines traditional STEM subjects such as science and maths. Recent work has begun to consider several explanations jointly. Leighton and Speer (2026), for example, use rich data on incoming US college students and a Gelbach decomposition to assess the roles of ability, self-assessed traits, life goals, high-school coursework and family background in the gender gap in STEM majors. Even with this extensive set of characteristics, they explain only 39% of the gap, with self-assessed ability and high-school coursework accounting for most of the explained component.

This study contributes to this literature in two ways. First, we show that gender gaps in secondary school STEM participation vary widely across subjects. In particular, large gaps are concentrated in practical, technology-oriented subjects, such as technology, engineering, construction, and design and communication graphics. In contrast, gaps in traditional academic STEM subjects are small or reversed. These practical subjects are often overlooked despite their economic importance and direct pathways to high-demand, high-wage STEM-related careers. Taking these subjects has also been found to significantly predict the likelihood of applying for a STEM college degree (Delaney and Devereux 2019). Second, we examine how these gender gaps evolve across secondary school by tracing subject choices from lower secondary through upper secondary education. This allows us to analyse how early subject choices persist and shape later educational pathways, consistent with evidence that prior exposure to a field can influence subsequent specialisation (Fricke et al. 2018).

We first document substantial gender gaps in enrolment in practical STEM subjects at the upper secondary level. These gaps are strongly related to earlier subject choices: participation in STEM subjects at lower secondary school, which we interpret as a measure of "STEM-readiness", explains much of the

[1] Examples of government policies to promote STEM participation among women include the USA's INSPIRE Women Act (2017) and Australia's Advancing Women in STEM Strategy.

later disparity. We therefore turn our attention to the lower secondary level and show that gender gaps are already pronounced at this early stage. These differences persist even after controlling for prior academic performance and a rich set of observable characteristics measured at age 9, indicating that the divergence in STEM pathways cannot be explained by the early-life factors we observe. Conventional explanations for gender gaps in STEM, including comparative advantage, school environments, and peer composition, account for only a small portion of the observed disparity in practical STEM subjects. We also show that the results are not driven by differences in subject availability across schools, an often-cited explanation for gender gaps in STEM, nor by gender sorting across schools; in fact, the gap is larger within coeducational schools than in single-sex schools. Finally, to ensure the findings are not driven by functional-form assumptions or variable selection, we conduct a robustness exercise using flexible machine learning methods and a substantially expanded set of explanatory variables. This approach captures rich nonlinearities and high-dimensional interactions, giving the model access to a much broader set of potential predictors. The results remain largely unchanged, increasing confidence that the early divergence in practical STEM participation is not explained by the observable characteristics available in our data.

The rest of the paper is organised as follows: Section 2 outlines the Irish education system, the Growing Up in Ireland data, and our methodological approach. Section 3 presents our results for upper secondary STEM enrolment, including regression analysis and decomposition of the gender gap. Section 4 turns to lower secondary subject choices and applies the same analytical framework. Section 5 addresses robustness, focusing on the potential role of school choice in driving results, particularly given the prevalence of single-sex schools in Ireland, and on the use of flexible machine learning methods to ensure that restrictive modelling assumptions do not drive the persistence of large unexplained gaps. Section 6 concludes.

2 Institutional Background, Data and Methodology

This research draws on data from the Growing Up in Ireland (GUI) Cohort '98, a nationally representative longitudinal study tracking the lives of Irish children. The study's first wave surveyed 8,568 children aged 9 from 910 randomly selected primary schools between August 2007 and May 2008. The sample accounts for roughly 14% of Ireland's nine-year-old population at the time. Data collection involved structured interviews with the children, their parents or primary caregivers, teachers, and principals. Unlike administrative data, the GUI dataset contains rich information on students' backgrounds and subject choices across schooling levels, making it well-suited to examining the early determinants of STEM subject selection.

Secondary education in Ireland begins around age 12 and is divided into two distinct phases, each ending with a national examination. The first phase, known as the Junior Cycle (i.e. lower secondary), spans three years and concludes with the Junior Certificate (JC). Students then progress to the Senior Cycle (i.e. upper secondary) around age 15, which lasts two to three years and culminates in the Leaving Certificate (LC), the primary qualification for entry to higher education. For context, the Junior Cycle is roughly equivalent to middle school in the U.S. education system, while the Senior Cycle corresponds to high school. On average, students take ten to eleven subjects for the JC and seven to eight for the LC. Throughout both phases, all students must study Irish, English, and Maths, with occasional exemptions from Irish granted to students with learning disabilities or those who relocated to Ireland after age 12. Schools determine the remaining subject choices based on availability and student interest. Maths and Irish are offered at higher, ordinary, or foundation levels, while English and all non-compulsory subjects are available at higher or ordinary levels. JC grades are letter-based, whereas LC grades are converted to a common points scale used to determine admission to post-secondary courses. Points are calculated from a student's best six subjects, with higher-level grades worth substantially more than ordinary-level grades. Since 2012, students who pass higher-level maths also receive a 25-point bonus.

The Junior Cycle offers a broad curriculum with limited subject choices. The Senior Cycle offers greater specialisation through a wider range of subjects. For instance, the general science subject studied in the Junior Cycle becomes specialised in physics, chemistry, and biology in the Senior Cycle. Students

retain the autonomy to choose their examination level for each subject. Access to most higher education courses is determined solely by LC performance. Third-level institutions commonly set subject-specific entry criteria; for example, Economics (BSc) at University College Dublin requires a Higher-Level grade 5 in Maths and an Ordinary-Level grade 6 in English. Among applicants who meet the subject-level and grade requirements, the highest-scoring individuals on the common points scale gain entry. Consequently, the subjects students choose in secondary school play a critical role in shaping access to higher education, as they determine whether students meet course prerequisites and how many points they can accumulate for admission to college programmes.

Ireland provides a particularly informative setting for studying gender differences in STEM subject choice, due to the continued prevalence of single-sex secondary schools. Single-sex schools remain a prominent feature of the Irish education landscape, with around 50% of students enrolled. Unlike in many other countries, these schools do not typically serve demographically distinct populations; their continued prevalence reflects historical legacy rather than deliberate sorting. Most have retained their original structures since their founding, when single-sex education was the norm.

2.1 Independent Variables

Recognising that secondary subject choices likely reflect multiple underlying factors, we exploit the unusually rich covariate information in the Growing Up in Ireland (GUI) survey to construct detailed measures of the main mechanisms proposed in the literature on gender gaps in STEM participation. Our approach draws on the extensive literature that primarily examines gender and higher education STEM enrolment, from which we borrow several theoretical frameworks (see Delaney and Devereux (2021) for a review of this literature and the potential theoretical mechanisms).

Specifically, we examine four distinct theoretical mechanisms highlighted as explaining gender gaps in STEM enrolment, and we also include a group of background controls. The first mechanism concerns "STEM-readiness". We examine upper secondary STEM subject choices because these choices directly influence STEM-readiness in higher education. However, existing gender gaps at the upper secondary level may already reflect earlier differences in STEM-readiness, potentially stemming from subjects studied at the lower secondary level. To address this, we include indicators for each optional JC subject chosen, indicators for whether students took mandatory subjects (maths and English) at a higher level, and an indicator for whether the student completed Transition Year (an optional one-year programme with a focus on personal and social development taken after the JC, before proceeding to the Senior Cycle).

The second mechanism concerns gender differences in skills, which may drive the gap either directly or through sex-specific comparative advantage, whereby students specialise in the subjects in which they are relatively strongest. Evidence consistently shows that female students outperform male students in reading, while males have a relatively small advantage in maths. Although the sources of these differences (biological, cultural, or stereotype-driven) are debated, their existence frequently explains part of the STEM gender gap (Delaney and Devereux 2019; Speer 2017). Thus, we incorporate multiple cognitive and non-cognitive skill measures from the GUI dataset. We measure cognitive skills at age 9 using standardised Drumcondra Verbal and Numerical Reasoning test scores. These standardised tests are administered nationally to students at multiple points throughout primary education. These assessments align with the Irish educational curriculum and evaluate students' proficiency in English (verbal reasoning) and Maths (numerical ability). The Drumcondra scores are also strongly associated with later academic attainment. In our sample, the correlations between Drumcondra reading scores and JC English performance and between Drumcondra numerical scores and JC maths performance are 0.464 and 0.513, respectively. These scores therefore provide a credible measure of academic ability prior to secondary school. We also include JC examination results, particularly grades in maths and English and the number of JC subjects taken at higher level, as direct proxies for academic ability.[2] One key strength of our data is that we can also include measures

[2]Following Doris et al. (2022), we convert JC grades and levels into a numerical measure, with higher scores reflecting stronger performance and higher examination level. The maximum score a student can attain in any subject is 12 and the minimum is 1.

of non-cognitive skills, which have been shown to explain gender disparities in educational choices at the secondary level (Buser et al. 2017). In particular, we include scores from the Strengths and Difficulties Questionnaire (SDQ), a measure of conscientiousness from the Ten Item Personality Inventory, and a measure of academic self-concept from the Piers-Harris scale.[3] The latter is essentially a measure of a student's own confidence in their academic ability, which has previously been proposed as a potential mechanism underlying gender gaps in achievement in STEM (Di Tommaso et al. 2021).

Third, we consider school factors and peer effects. A common focus of the literature is the effect of single-sex schooling on the propensity of female students to take STEM courses (Calkins et al. 2023). Our study's Irish context is particularly interesting given the high prevalence of single-sex secondary schools. While our data do not permit direct identification of peer effects, we include extensive information about school characteristics that could influence STEM enrolment decisions. School characteristics include subject offerings; a key advantage of our data is that we observe the full set of subjects offered by each school, allowing us to control directly for subject availability, a factor frequently highlighted in the literature on gender gaps in STEM. Other school-level controls include school type (e.g., fee-paying or non-fee-paying) and whether the school is designated as a "DEIS" school (i.e., one that serves a high proportion of students at risk of educational disadvantage). To further approximate peer influences, we explicitly include indicators of single-sex versus mixed-gender school environments at the secondary level.

Fourth, gender gaps in STEM enrolment may reflect fundamental differences in preferences regarding fields of study or related career paths. For instance, Kuhn and Wolter (2022) highlight that female students tend to favour fields involving interpersonal interactions, a description less applicable to many practical STEM subjects. Such differences in preferences may stem from societal norms or gender stereotypes. While we do not directly observe preferences for specific job characteristics, the dataset contains rich information on students' interests, attitudes, and aspirations. In particular, we observe students' stated occupational aspirations, which we categorise by broad employment field. We also incorporate measures of students' enjoyment of maths and reading, whether their stated favourite subject in the JC is practical, and whether they report liking school. Finally, we account for students' perceptions of the difficulty of maths and English, and their expectations regarding their own educational attainment (how far they expect to go in their educational journey).

In addition to these four mechanism groups, we incorporate a range of background variables. While these are not expected to explain gender gaps directly, they may nevertheless influence overall STEM enrolment and, if omitted, could confound the analysis.[4] These variables include the primary caregiver's immigration status (born in Ireland or not) and education (university-level educated or not), the number of books in the home, and health-related measures, such as whether the young person has a disability.

Starting with an initial balanced panel of 6,039 students observed across Waves 1 to 3, the sample size is reduced mainly because of missing data on key variables central to the analysis. In particular, we exclude students who did not report their JC maths or English level or grade, as well as students attending schools for which the principal did not complete the school characteristics survey in Waves 2 or 3. These exclusions account for the majority of the reduction in sample size and are necessary given the paper's core focus on subject-level outcomes and school characteristics. After applying all sample restrictions, the final analytical sample consists of 3,888 students.[5] The median student attends a school with 14 students in our data; 11.4% attend schools with five or fewer.[6] In total, we use 58 independent variables, which we organise into the four mechanisms listed above plus background characteristics: STEM-readiness, skills/comparative advantage, school factors/peer effects, preferences/tastes, and background characteristics. We select these

[3]The Strengths and Difficulties Questionnaire (SDQ) serves as a behavioural screening instrument. The primary caregiver (over 99% of PCGs are recorded as the child's mother) completes the questionnaire, which assesses 25 attributes related to the child and covers topics such as emotional, conduct, and prosocial behaviour. The questionnaire yields a total score, with higher scores indicating greater behavioural and emotional difficulties.

[4]Related evidence using GUI data suggests that family background may be associated with STEM field choice differently by gender. In particular, Crotty (2025) finds that females with postgraduate-educated mothers are significantly less likely to choose a STEM major in higher education.

[5]See Appendix Table 10 for a breakdown of observations lost through our sampling restrictions.

[6]See Appendix Table 11 for full details on the distribution of students per school.

core variables ex ante because they map transparently onto the main mechanisms emphasised in the literature and can therefore be grouped in an interpretable Gelbach decomposition. This comes at the cost of excluding some additional variables that may contain predictive information but do not fit cleanly into these mechanism groups, are measured less directly, or would substantially reduce the transparency of the main specification. We use these additional variables in the machine learning analysis in Section 5.1. Table 1 provides summary statistics for a subsample of variables from each grouping. Appendix Section A.3 lists all variables in this analysis. Crotty and Murphy (2026) use a similar dataset and variable set.

2.2 Dependent Variable

The optional LC STEM subjects fall into four categories: science subjects (Biology, Chemistry, Physics, Agricultural Science), technology subjects (Technology, Design and Communication Graphics (DCG)), engineering subjects (Engineering, Construction Studies), and maths subjects (Maths, Applied Maths). Two other STEM subjects currently on the LC curriculum are excluded from our analysis: Computer Science, which was not offered as a subject when our sample sat their LC exams, and a combined "Physics & Chemistry" subject, which is offered by few schools and therefore taken by an extremely low number of students in our sample and nationally. The STEM subjects listed above vary markedly from each other. Technology, DCG, Construction, Engineering and Agricultural Science feature substantial practical components (for example, designing and constructing a product for Engineering or submitting a report on practical farm experience for Agricultural Science). Meanwhile, the other subjects (Maths, Applied Maths, Biology, Chemistry, and Physics) are more traditional academic subjects and are assessed solely by a written final exam. As a result, one might expect different factors to influence enrolment in these two subject groups.

Table 1 Summary Statistics - Independent Variables

Variable	Mean	Max	Min
STEM-Readiness			
Studied Technical Graphics for JC	0.20	1.00	0.00
Studied Metalwork for JC	0.10	1.00	0.00
Studied Music for JC	0.20	1.00	0.00
Studied Foreign Language for JC	0.90	1.00	0.00
Skills/Comparative Advantage			
Drumcondra Reading Score	0.12	2.87	-3.36
Drumcondra Numerical Score	-0.65	1.90	-3.62
Number of Higher-Level Subjects Taken for JC	8.21	12.00	0.00
Academic Self-Concept Score	12.10	16.00	0.00
School Factors/Peer Effects			
Attended DEIS School	0.15	1.00	0.00
Attended Fee-paying School	0.11	1.00	0.00
Agricultural Science Offered by School	0.66	1.00	0.00
Attended All Boys Secondary School	0.23	1.00	0.00
Preferences/Tastes			
Favourite Subject for JC is Practical Subject	0.18	1.00	0.00
Young Person (YP) Likes Maths	1.61	3.00	1.00
YP Likes School	1.81	3.00	1.00
YP Preferred adult occupation – Healthcare	0.11	1.00	0.00
Background Characteristics			
Primary Caregiver (PCG) is University Educated	0.18	1.00	0.00
YP has a Disability	0.10	1.00	0.00
Number of Books in home for YP	4.25	5.00	1.00
PCG born outside of Ireland	0.15	1.00	0.00

Notes: N=3,888. YP = young person (survey child); PCG = primary caregiver (typically mother). All statistics are computed using survey weights. Binary variables are coded as indicators taking values 0/1. Ordinal variables ("Likes Maths", "Likes School") are measured on a 1–3 scale, 1=Always, 2=Sometimes, 3=Never. "Number of Books in Home" is measured on a 1–5 categorical scale. A full list of variables is provided in Appendix Table 13.

Table 2 presents the percentage of students of each gender enrolled in each subject of interest in our sample. Appendix Section A.2 provides national-level enrolment rates to show that our data are nationally representative. Clear gender differences emerge in subject selection patterns. Despite widespread discourse highlighting the underrepresentation of females in STEM, Biology and Chemistry have higher enrolment rates among female students than male students. While gender gaps favouring male students are also evident in Physics and Applied Maths, these are relatively modest compared to the substantial disparities observed in practical STEM subjects. Using the same dataset, Crotty and Murphy (2026) show substantial disparities in enrolment rates in academic STEM subjects along socioeconomic lines, not gender. Large gender gaps favouring male students are observed in construction studies (22.22% of boys versus 1.82% of girls), design and communication graphics (DCG) (15.57% of boys versus 3.57% of girls), and engineering (14.21% of boys versus 0.74% of girls). The gender gap in agricultural science is smaller but still substantial (14.26% of boys versus 9.42% of girls). Technology has the lowest overall enrolment rate in our sample, with 3.38% of students in the sample taking the subject, but there is nevertheless a substantial gender gap (5.58% of boys versus 1.06% of girls).

Despite their prevalence in the curriculum, studies of STEM gender gaps often overlook practical STEM subjects. However, recent evidence underscores their importance. Using administrative data, Delaney and Devereux (2019) show that, among college applicants, taking any of these five practical subjects significantly increases the probability of enrolling in a STEM course in higher education in Ireland. Delaney and Devereux (2019) find that all five of these subjects have a statistically significant positive effect on the likelihood of applying for a STEM degree. Strikingly, four of the five practical subjects (Technology, DCG, Engineering, and Construction Studies) predict STEM enrolment better than Biology, the most popular STEM subject at LC level. Only Chemistry, Physics, and the rarely taken combined Physics

& Chemistry subject outperform Engineering, Technology and DCG as predictors of STEM enrolment in college. This evidence highlights the economic relevance of understanding who takes practical STEM subjects. Given their importance as predictors of post-secondary STEM enrolment, the substantial gender gaps in participation, and the limited existing research, we focus our analysis on practical STEM subjects. Below is a brief description of these subjects.[7]

Both technology subjects place strong emphasis on the use of information technology. Design and Communication Graphics (DCG) emphasises spatial reasoning, graphical communication, and CAD (computer-aided design). Core areas of study include plane and descriptive geometry and information and communication technologies. Technology focuses on solving technological problems, with core topics including information and communication technologies, energy, electricity and electronics, and communications and graphic media. The Construction syllabus explicitly aims to provide a basis for those who want to study construction technologies in third-level education, and it therefore combines theoretical knowledge of construction principles with practical workshop and laboratory skills. Similarly, engineering studies a wide range of mechanical engineering materials, processes and technological applications. Agricultural science is the study of the scientific and technological principles and practices underlying agriculture. Topics include plant and animal biology, soil science, and sustainability, emphasising experiential learning via the submission of a "Practical Experience Report." While the specific subject titles are unique to the Irish curriculum, these subjects closely correspond to applied STEM subjects offered in many other education systems.

To incorporate all of the practical subjects listed above, we construct a composite outcome variable called *Practical STEM*, which equals 1 if a student studies at least one of these subjects and equals 0 otherwise.[8] 45.96% of the 1,906 male students in our sample take at least one practical STEM subject at Leaving Certificate level (and therefore have a value $Practical\ STEM = 1$), compared to 15.63% of the 1,982 female students. This gives us a gender gap of 30.33 percentage points. Throughout the paper, we calculate gender gaps as the difference in weighted mean enrolment rates between male and female students, with a positive gender gap indicating that a lower proportion of female students enrol than male students. We use sampling weights throughout our analysis to address concerns about selective attrition in our survey data. All reported enrolment rates and regression/decomposition estimates use survey weights, and running the analysis without weights yields similar results. These gaps far outweigh those observed in traditional academic STEM subjects, which the existing literature tends to focus on. If we define a similar binary variable *Academic STEM*, which equals 1 if a student takes at least one of biology, chemistry, physics, or applied maths and 0 otherwise, the gender gap reverses in favour of girls at -6.41%.

[7] See https://www.curriculumonline.ie/ for further information on each subject's curriculum.

[8] In addition to the aggregate practical STEM outcomes, we also examine gender gaps separately for individual subjects. The corresponding results are reported in Appendix Tables 17 and 18.

Table 2 LC STEM Enrolment by Gender

Subject	Male (%)	Female (%)	Total (%)
Academic STEM Subjects			
Biology	48.88	69.03	58.70
Chemistry	15.89	20.09	17.94
Physics	24.94	8.66	17.01
Applied Maths	8.07	3.11	5.65
Academic STEM (1+ academic STEM subject)	70.46	76.86	73.58
Practical STEM Subjects			
Construction Studies	22.22	1.82	12.28
Agricultural Science	14.26	9.42	11.90
Design and Communication Graphics	15.57	3.57	9.72
Engineering	14.21	0.74	7.64
Technology	5.58	1.06	3.38
Practical STEM (1+ practical subject)	45.96	15.63	31.17
Total Observations	1,906	1,982	3,888

Notes: Percentages report weighted enrolment rates by gender for Leaving Certificate (LC) STEM subjects; for example, 48.88% of male students, 69.03% of female students, and 58.70% of all students in the sample study biology for LC. Composite rows ("Practical STEM" and "Academic STEM") indicate whether a student is enrolled in at least one subject within the respective category.

2.3 Methodology

We implement the decomposition within a linear probability model (LPM) framework. Let i index students and let Practical STEM$_i$ denote an indicator for whether student i chooses a Practical STEM subject at upper secondary level. We begin with a baseline specification that captures the raw gender gap:

$$\text{Practical STEM}_i = \alpha + \beta\,\text{Male}_i + \varepsilon_i, \tag{1}$$

where Male$_i$ is an indicator equal to one for male students. The coefficient β represents the unconditional gender gap in Practical STEM take-up. We then estimate a specification that includes a comprehensive set of covariates:

$$\text{Practical STEM}_i = \alpha + \beta^F\,\text{Male}_i + \mathbf{X}_i^{SR}\boldsymbol{\gamma}^{SR} + \mathbf{X}_i^{SP}\boldsymbol{\gamma}^{SP} + \mathbf{X}_i^{TP}\boldsymbol{\gamma}^{TP} + \mathbf{X}_i^{CA}\boldsymbol{\gamma}^{CA} + \mathbf{X}_i^{BG}\boldsymbol{\gamma}^{BG} + \varepsilon_i, \tag{2}$$

where $\mathbf{X}_i^{SR}$ captures STEM-readiness, $\mathbf{X}_i^{SP}$ captures school factors/peer effects, $\mathbf{X}_i^{TP}$ includes measures of preferences/tastes, $\mathbf{X}_i^{CA}$ captures skills/comparative advantage, and $\mathbf{X}_i^{BG}$ contains background characteristics. The coefficient β^F therefore represents the conditional gender gap after accounting for these observable characteristics, where the superscript F denotes the full-specification estimate. All models are estimated using ordinary least squares, and standard errors are clustered at the school level to account for within-school correlation in outcomes.

To quantify the contribution of each group of covariates to the reduction in the gender coefficient between equations (1) and (2), we apply the decomposition proposed by Gelbach (2016). This approach exploits the omitted variable bias formula to express the change in the gender coefficient as the sum of contributions from each covariate group:

$$\beta - \beta^F = \Delta^{SR} + \Delta^{SP} + \Delta^{TP} + \Delta^{CA} + \Delta^{BG}. \tag{3}$$

Each Δ^g represents the portion of the change in the gender coefficient attributable to covariate group g. The Gelbach decomposition provides a nested and path-independent way of allocating the reduction in the gender gap across groups of explanatory variables.

3 Results - Upper Secondary

This section analyses enrolment in practical STEM subjects at the LC level. The analysis proceeds in two steps. First, we present regression results that describe which factors are associated with practical STEM take-up at upper secondary school. Second, we focus explicitly on gender differences: we quantify the unconditional gender gap, examine how it changes when conditioning on the full set of covariates, and apply a Gelbach decomposition to attribute changes in the gender coefficient to groups of covariates.

3.1 Upper Secondary Regression Results

We begin by estimating equation (2) for enrolment in practical STEM subjects at the LC level. The regression includes grouped controls capturing STEM readiness, skills/comparative advantage, preferences/tastes, school factors/peer effects, and background characteristics. Appendix Table 15 reports coefficient estimates for the full specification.

The results indicate that enrolment in practical STEM subjects at LC level is strongly associated with prior curricular exposure. In particular, students who studied practical subjects at lower secondary level (Material Technology, Metalwork, Technical Graphics, and Technology) are significantly more likely to enrol in practical STEM subjects at upper secondary. In addition, students who took more higher-level subjects during the Junior Cycle are more likely to participate in practical STEM later. These findings point to the importance of early curricular pathways. Students who select into practical or technically oriented subjects earlier in secondary school are substantially more likely to continue along a similar trajectory at LC level. This suggests that participation in practical STEM at upper secondary largely continues pathways established during lower secondary education. By the time students make their LC subject choices, a considerable portion of the divergence in participation has already emerged through earlier subject selection and exposure.

Several other factors also relate to practical STEM participation. Students whose favourite JC subject is a practical subject are significantly more likely to enrol in practical STEM, as are those who report occupational aspirations in transport- or agriculture-related fields. School-level subject availability also matters: attending a school that offers agricultural science or biology is positively associated with practical STEM enrolment. In contrast, students who dislike maths are less likely to take practical STEM subjects than those who always like maths. Similarly, students whose primary caregiver was born outside Ireland are less likely to participate. Finally, higher reading scores at age nine are negatively associated with practical STEM enrolment, consistent with a comparative advantage mechanism whereby students with stronger verbal skills are less likely to specialise in practical or technical subjects.

3.2 Upper Secondary Gender Gap Decomposition

We now turn to gender differences in practical STEM enrolment in Table 3. The unconditional gender gap is large: 45.96% of male students take at least one practical STEM subject, compared with 15.63% of female students, for a gap of 30.3 percentage points. This raw gap corresponds to the coefficient on the male indicator in a baseline specification that includes only gender. Conditioning on the full set of covariates reduces the estimated male coefficient to 7.1 percentage points. Observable differences captured by the covariate set therefore explain a substantial proportion of the unconditional disparity.

To understand which mechanisms account for the reduction in the male coefficient from the baseline to the full model, we apply the Gelbach (2016) decomposition. Panel B of Table 3 reports the decomposition results, showing the extent to which each covariate group accounts for the change.

STEM-readiness, captured by JC subject choices and related indicators, makes the dominant contribution. This group accounts for 17.4 percentage points of the reduction in the gender coefficient (17.4/30.3 = 57.4% of the total gap). This implies that a large share of the gender gap observed at the end of secondary school reflects earlier differences in curricular pathways established by lower secondary subject

choices, typically made around age 13. These results reinforce a pipeline interpretation, mirroring the well-established finding that access to and enrolment in STEM in higher education depend on preparation accumulated during secondary school. Our contribution is to show that the same logic applies one step earlier: the STEM-readiness that matters for higher education is itself shaped by choices made at the start of secondary school. Many students, particularly female students, may have already limited their future STEM opportunities well before reaching the senior cycle.

Preferences/tastes explain an additional 5.8 percentage points (19.2% of the total gap). Skills/comparative advantage, school factors/peer effects, and background characteristics contribute relatively little to the change in the male coefficient. This suggests that gender differences in practical STEM participation at the LC level are not primarily a story of girls lacking the measured skills needed to succeed, nor a story driven by socioeconomic background. Rather, gender differences at the end of secondary school are largely explained by earlier curricular sorting and persistence. Once students have entered different subject pathways, subsequent choices reflect that accumulated exposure.

Table 3 LC Practical STEM — Gender gap and Gelbach decomposition

Panel A: OLS estimates		
Baseline - Male (1)	0.303***	(0.023)
Full Model - Male (2)	0.071**	(0.034)
Observations	3,888	
Panel B: Gelbach decomposition of the change in the gender gap		
Total Change: (1)–(2)	0.233***	(0.033)
Contribution to change:		
STEM-Readiness	0.174***	(0.023)
Skills/Comparative Advantage	0.0004	(0.005)
School Factors/Peer Effects	0.0001	(0.018)
Preferences/Tastes	0.058***	(0.014)
Background Characteristics	-0.0005	(0.003)

Note: Panel A reports the coefficient on the male indicator from (i) a baseline specification including only gender and (ii) a full specification including controls for STEM-readiness, skills/comparative advantage, preferences/tastes, school factors/peer effects, and background characteristics. Panel B reports a Gelbach (2016) decomposition of the difference in the Male coefficient between the baseline and full specification, allocating the total change across pre-defined covariate groups. The dependent variable is an indicator equal to one if the student takes at least one practical STEM subject at Leaving Certificate level and zero otherwise. Coefficients and decomposition contributions are expressed as proportions (e.g. 0.303 corresponds to 30.3 percentage points). Robust standard errors clustered by school in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4 Results - Lower Secondary

Given that enrolment in practical STEM subjects at the lower secondary level is the strongest predictor of enrolment in similar subjects at the upper secondary level, a natural next step is to examine the determinants of these earlier choices. We therefore model students' decisions to enrol in practical STEM subjects at the lower secondary stage. Our analysis relies on explanatory variables measured in Wave 1, when students were aged nine and had not yet entered secondary school or made subject choices. These variables capture a broad set of early-life characteristics, including measures of cognitive and non-cognitive skills and proxies for students' interests, aspirations, and preferences, allowing us to address theories of comparative advantage and individual preferences. Because we measure the explanatory variables before

secondary school entry, we do not include secondary school characteristics in this specification. We aim to isolate factors that shape subject choices before students select into particular schools or are exposed to secondary school environments. We therefore examine the role of school environments separately in subsequent analyses and robustness checks. Similarly, we do not include a measure of STEM-readiness in the lower secondary decomposition. While STEM-readiness is an important predictor of STEM participation in upper secondary and higher education, variation in STEM-readiness before secondary school is essentially zero, as all students follow the same national primary school curriculum. STEM-readiness can therefore be discounted as a potential driver of the gender gap at lower secondary.

As in the upper secondary analysis, we proceed in two steps. First, we present regression results describing the correlates of enrolment in practical STEM subjects at the JC level. Second, we quantify the gender gap at this stage, show how it changes when conditioning on observables measured at age 9, and apply a Gelbach decomposition to attribute the change in the gender coefficient between the baseline and full specifications to groups of covariates.

4.1 Lower Secondary - Summary Statistics

Students in our sample could take four practical STEM subjects at the lower secondary level: Material Technology (MT), Technical Graphics (TG), Metalwork, and Technology. The decomposition results from the previous section suggest that enrolment in practical STEM subjects at the lower secondary level is a key driver of participation in these subjects at the upper secondary level, which in turn influences college STEM enrolment. Each subject can be considered a "feeder" subject for a corresponding upper-secondary subject. 79% of students in our sample who take Construction at Leaving Certificate had taken Material Technology at Junior Certificate, 76% who take DCG at Leaving Certificate had taken TG at Junior Certificate, 68% who take Engineering had taken Metalwork at Junior Certificate, and 54% who take Technology at Leaving Certificate had taken Technology at Junior Certificate.[9] Table 4 shows the proportion of each gender taking each subject. Enrolment rates for each STEM subject at the national level show that the sample is representative of individuals taking the JC in Ireland during the sample period. See Appendix Section A.2 for these statistics.

To ensure consistency between our lower secondary and upper secondary analyses, we define the composite outcome variable *JC Practical STEM* as 1 if a student studies at least one of the four practical subjects and 0 otherwise. The gender gap is even larger at this stage than at the LC level: 72.55% of male students have *JC Practical STEM* = 1, compared with 13.78% of female students, a difference of 58.77 percentage points.

Table 4 JC Practical Enrolment by Gender

Subject	Male (%)	Female (%)	Total (%)
Practical STEM Subjects			
Material Technology	41.57	5.54	24.01
Technical Graphics	33.25	5.93	19.93
Metalwork	18.12	1.83	10.18
Technology	10.97	1.96	6.58
JC Practical STEM (1+ practical subject)	72.55	13.78	43.91
Total Observations	1,906	1,982	3,888

Notes: Percentages report weighted enrolment rates by gender for Junior Certificate (JC) Practical STEM subjects; for example, 41.57% of male students, 5.54% of female students, and 24.01% of all students in the sample study material technology for JC. The composite row ("JC Practical STEM") indicates whether a student is enrolled in at least one of these subjects.

[9] Further information on each subject's curriculum can be found at https://www.gov.ie/en/department-of-education/collections/junior-certificate-syllabuses-and-prescribed-material/.

4.2 Lower Secondary Regression Results

The main specification regresses JC Practical STEM on a rich set of covariates measured at age 9. Appendix Table 16 reports the estimates from the fully specified model of this education stage. The estimated gender coefficient is large and highly significant. Meanwhile, most other covariates are statistically insignificant, suggesting that early-life characteristics explain little variation in practical STEM participation at this stage.

However, there are exceptions: students with a lower Drumcondra reading score at age 9 are more likely to enrol in practical STEM subjects, consistent with a comparative advantage mechanism in which students with relatively weaker verbal skills are more likely to specialise in practical or technical subjects. There is also weak evidence that students who report never liking school are more likely to enrol in practical STEM subjects, although this effect is only marginally statistically significant. Students who aspire to a career in a technical or trade/manual sector are more likely to enrol in practical STEM subjects, highlighting the role of early career preferences in shaping subject choices, even at the lower secondary level. Finally, students whose primary caregiver does not have a university degree are more likely to enrol in practical subjects, suggesting that socioeconomic background is associated with selection into different curricular pathways.

4.3 Lower Secondary Gender Gap Decomposition

The unconditional gender gap in *JC Practical STEM* is 58.77 percentage points, corresponding to the male coefficient in a baseline model that includes only a gender indicator. Conditioning on the full set of age-9 covariates reduces the male coefficient only slightly, from 58.77 to 56.98, a 1.79 percentage-point reduction. Thus, only 3% of the lower secondary gender gap is explained by the observable characteristics available in our data before secondary school entry.

Table 5 reports the results of the Gelbach decomposition. Preferences/tastes account for the largest share of the explained component, but their contribution is modest in absolute terms, reducing the male coefficient by 2.21 percentage points. Skills/comparative advantage and background characteristics contribute negligibly, with effects close to zero. The vast majority of the disparity remains unexplained.

The lower secondary results imply that a wide range of differences observable in late primary school cannot account for the practical STEM gender gap. The GUI data contain unusually rich measures of cognitive skills, non-cognitive traits, academic self-concept, early interests, occupational aspirations and family background. Yet, despite this breadth of information, conditioning on these measures explains only a trivial share of the JC gap. This echoes the difficulty of explaining gender differences at later stages of the STEM pipeline. Using detailed measures of ability, self-assessed traits, life goals, high-school coursework and family background, Leighton and Speer (2026) explain 39% of the gender gap in STEM majors among incoming US college students. Our results locate an even larger unexplained component considerably earlier: characteristics measured before secondary school explain only 3% of the gender gap that emerges in practical STEM subject choice by age 13. The gap does not appear to be driven primarily by measured differences in preferences, family background, comparative advantage, or STEM-readiness.

The large unexplained component admits two interpretations. First, the gap may be driven by gender norms and socialisation that standard survey measures capture poorly, such as beliefs about which subjects are "for boys" or "for girls", the anticipated social cost of choosing a gender-atypical subject, or differential encouragement from parents, teachers, or peers. Second, the divergence may emerge largely between ages 9 and 13, during late primary school and the transition into adolescence, and therefore after we measure our covariates. In either case, the unexplained component should not be read as an absence of mechanism, but as evidence that the mechanisms most commonly emphasised in the literature do not account for the gap in our setting.

Our results are also inconsistent with a simple story in which girls at age 9 already report different interests and aspirations that map onto later subject choices, as these measures contribute little to explaining the gap. If gendered attitudes are a key driver, these early self-reports either do not capture them or they have not yet formed by age 9.

Considering each theoretical explanation individually provides further clarity. First, as noted above, STEM-readiness is not a relevant mechanism at the lower secondary level, so that it can be ruled out as a potential explanation for the gender gap. Second, the comparative advantage theory, as captured by cognitive assessments, explains only a negligible proportion of the gender gap. This suggests that skill differences observable at age 9 play a minimal role in shaping later practical STEM subject choices. In particular, the gap does not appear to reflect girls' stronger academic ability drawing them towards more traditional academic subjects.

Third, preference-based theories, proxied by students' early reported feelings towards maths, reading and school in general, as well as their occupational aspirations, contribute slightly more, but still explain only a small portion of the gender disparity (3.76% of the total gap). While our data allow for a broad approximation of early preferences, capturing every relevant dimension of student interests and aspirations at this age is nearly impossible. As such, we cannot rule out that underlying differences in preferences, potentially shaped by early socialisation or gender norms, drive the observed gap. This finding suggests that the gender gap is not driven by measured differences in educational or career aspirations, or differences in general interests or enjoyment of core subjects. Still, it may instead reflect attitudes or stereotypes we do not directly measure.

Fourth, background characteristics, including socioeconomic indicators and family conditions, appear to have no meaningful explanatory power for the gender gap. This aligns with expectations, as background variables, while potentially influential on overall enrolment decisions, are, on average, the same for male and female students. Lastly, because the lower secondary analysis omits school-level factors, a deeper examination of school and peer effects at later stages of education is warranted. We return to the role of schools in Section 5.2.

Overall, the overwhelming dominance of the residual gender component in this early-stage decomposition underscores the importance of exploring intermediate educational stages more closely, with particular attention to socialisation, gender norms and teacher interactions. Targeted policy interventions and mentorship programmes designed to counter these factors before secondary school may be vital for mitigating the substantial gender gap observed in practical STEM subjects at both lower and upper secondary levels, and the subsequent gender gaps in college.

5 Robustness Testing

5.1 Machine Learning

The preceding results show that while a large share of the gender gap in LC practical STEM enrolment can be traced back to JC subject choices, the (even larger) gender gap in JC practical STEM enrolment is almost entirely unexplained by the rich set of characteristics we observe at age 9. One concern is that this finding could be an artefact of our modelling choices: the linear probability model imposes strong functional-form restrictions, and our decomposition relies on a pre-specified set of covariates grouped into broad mechanisms. To assess whether the unexplained component is driven by these parametric assumptions or by variable selection, we complement the regression-based analysis with a flexible machine-learning approach that allows for rich nonlinearities and interactions, and lets the data determine which variables are most predictive.

Table 5 JC Practical STEM — Gender gap and Gelbach decomposition

Panel A: OLS estimates		
Baseline - Male (1)	0.588***	(0.034)
Full Model - Male (2)	0.570***	(0.032)
Observations	3,888	
Panel B: Gelbach decomposition of the change in the gender gap		
Total Change: (1)–(2)	0.018*	(0.011)
Contribution to change:		
Skills/Comparative Advantage	-0.002	(0.004)
Preferences/Tastes	0.022**	(0.010)
Background Characteristics	-0.002	(0.004)

Note: Panel A reports the coefficient on the male indicator from (i) a baseline specification including only gender and (ii) a full specification including controls for skills/comparative advantage, preferences/tastes, and background characteristics. Panel B reports a Gelbach (2016) decomposition of the difference in the Male coefficient between the baseline and full specification, allocating the total change across pre-defined covariate groups. N=3,888. The dependent variable is an indicator equal to one if the student takes at least one practical STEM subject at Junior Certificate level and zero otherwise. All covariates are measured at age 9, before entry to secondary school. Coefficients and decomposition contributions are expressed as proportions. Robust standard errors clustered by school in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Specifically, we fit gradient-boosted decision tree (XGBoost) models to predict practical STEM participation at both LC and JC stages, using an expanded set of available covariates.[10] Our main specification includes covariates linked to theoretical mechanisms highlighted in the literature as potential explanations for gender gaps in STEM-based choices. This allows for an interpretable assessment of their contribution to the gender gap while avoiding overly broad controls in the main analysis. The machine learning analysis draws on a broader set of variables from the GUI data to capture additional patterns. We then decompose the predicted gender gap using SHAP (Shapley Additive exPlanations) values, which attribute each individual prediction to contributions from each feature. Aggregating SHAP values across individuals allows us to express the model-implied gender gap as the sum of contributions from our mechanism groups (STEM-readiness, skills/comparative advantage, preferences/tastes, school factors/peer effects, background characteristics), with the remaining component captured by the SHAP contribution of the gender indicator itself (i.e. the part not explained by other observed characteristics in the model). We provide full details on model tuning, train/test splits, and variable construction in Appendix Sections A.4 and A.5.

The machine-learning results in Table 6 closely mirror the regression-based conclusions. At the upper-secondary stage, the model predicts a gender gap in LC practical STEM participation similar to the observed gap, and the decomposition attributes a substantial share of this gap to differences in JC subject choices (STEM-readiness), with smaller roles for other covariate groups. At the lower secondary stage, the model again predicts a large gender gap in JC practical STEM participation. Still, the decomposition assigns only a minimal share of the gap to observed characteristics measured at age 9, with the vast majority attributed to the residual direct contribution of gender. The persistence of this pattern in a flexible, high-dimensional setting suggests that the large unexplained gender gap at the lower secondary level is unlikely to be driven by restrictive functional-form assumptions or by an overly narrow choice of explanatory variables. Moreover, the fact that moving to a substantially richer, more flexible set of predictors, including a broader set of child attitudes, behavioural measures, school experiences, and family background indicators (see Appendix Table 13), leaves the explained share essentially unchanged suggests

[10] In addition to the covariates used in the decomposition analysis, the machine learning models include an additional 66 variables capturing student skills, preferences, background characteristics, and school factors. These extra variables are reported in Appendix Table 13.

that the result is unlikely to be driven by age-9 characteristics available in the GUI survey but excluded from our baseline model. Instead, the evidence points to forces that emerge after age 9, during late primary school or the transition into adolescence, or to mechanisms that are not well captured by standard survey measures at that age.

Table 6 SHAP Decompositions of Gender Gaps in Practical STEM (LC and JC)

	LC Practical STEM SHAP	**JC Practical STEM SHAP**
Observed Gap (test data)	28.87	58.62
Predicted Gap (test data)	31.15	50.72
Contribution to predicted gap:		
STEM-Readiness	14.33	—
School Factors/Peer Effects	1.94	0.61
Preferences/Tastes	5.45	2.82
Skills/Comparative Advantage	1.00	0.63
Background Characteristics	0.17	0.20
Gender (residual)	8.26	46.46

Notes: Results from XGBoost models predicting practical STEM participation at Leaving Certificate (LC) and Junior Certificate (JC) levels. Models are estimated using a 70% training / 30% test split. "Observed gap" and "predicted gap" refer to weighted gender differences in mean outcomes in the test data, measured in percentage points. Contribution rows report each covariate group's contribution to the predicted gender gap, in percentage points, based on SHAP values aggregated over the test sample. "Gender (residual)" captures the residual contribution of the gender indicator not explained by other observed variables. "—" indicates that the covariate group is not included in the JC specification. The reported sample corresponds to the held-out test set.

5.2 The Role of School

Ireland is a particularly interesting case study when examining gender disparities in secondary school, as it has an unusually high prevalence of single-sex schools. If boys-only and girls-only schools tend to offer different subject ranges to their students, it is possible that gender differences in subject enrolment reflect school-level sorting rather than individual preferences or characteristics. As such, a potential concern is that the subject choices analysed in Sections 3 and 4 may be heavily influenced by a student's (or parent's) choice of secondary school. In this section, we examine whether our results are primarily driven by differences between schools or whether the gender gaps reflect deeper patterns rooted in students' decisions during lower secondary education.

We begin by examining variation in subject offerings across secondary schools of different gender compositions (boys-only, girls-only, and mixed) in Figure 1. Due to data restrictions from the data provider, we cannot report the proportions of girls-only schools offering Construction and Engineering in our sample at the LC level. Technology is rare in girls-only schools and uncommon in boys-only schools; DCG is rare in girls-only schools but widely offered in boys-only schools. The same pattern is observed at the Junior Certificate level; girls-only schools are less likely than boys-only schools to offer all of Material Technology, Technical Graphics, Metalwork and Technology, although all of these subjects are more commonly offered in mixed schools than in boys-only schools. As male and female students are systematically sorted into different schools that offer different subject sets, these disparities could be driving our results. Single-sex schools account for a large proportion of our sample, with roughly 26% of students attending girls-only schools and a further 23% attending boys-only schools. National statistics are provided for comparison in Appendix Table 9.

If school subject offerings were the primary driver of gender disparities in practical STEM enrolment, we would expect these gaps to diminish in coeducational settings where male and female students have

identical subject choice sets. However, as shown in Table 7, the reverse often holds. Across both the Junior and Leaving Certificate cycles, gender gaps in practical STEM participation are larger within mixed schools, suggesting that even when male and female students attend the same institution and have equivalent subject options, female students are far less likely to choose practical courses. The only exception to this finding is DCG, which shows a marginally larger gender gap within single-sex schools.

This finding contradicts the notion that single-sex schools are the primary source of gender imbalances in practical STEM uptake. Instead, it indicates that gender differences in subject preferences and choices persist, and may even intensify, in environments with equal access to the full subject choice set. This is consistent with Delaney and Devereux (2019), who find an approximately four percentage-point larger STEM gender gap in mixed-sex schools, and with evidence that exposure to same-sex peers reduces girls' likelihood of sorting into gender-typical fields (Schneeweis and Zweimüller 2012). One possible explanation is that gender norms are more salient in coeducational settings, where choosing a gender-atypical subject may carry a greater social cost. This finding is especially noteworthy because mixed schools represent the single largest category in our sample.

Table 7 Gender Gaps in Subject Enrolment by School Type

Subject	Gender Gap - Single-Sex (%)	Gender Gap - Mixed (%)
Leaving Certificate outcomes		
LC Practical STEM	25.49	33.58
Leaving Certificate subjects		
Construction Studies	13.42	25.80
Engineering	4.10	20.88
Design and Communication Graphics	12.12	11.72
Agricultural Science	4.47	4.80
Technology	3.78	5.19
Junior Certificate outcomes		
JC Practical STEM	52.90	62.45
Junior Certificate subjects		
Material Technology	26.01	43.50
Metalwork	4.58	25.46
Technical Graphics	26.74	27.38
Technology	7.02	10.62
Observations	1,900	1,988

Note: Gender gaps are calculated as the difference in weighted mean enrolment rates between male and female students (male minus female), expressed in percentage points. Positive values indicate higher enrolment among male students. "Single-sex schools" combine boys-only and girls-only schools; "mixed schools" include coeducational schools only. Estimates are computed separately within each subsample. The full sample includes 3,888 students.

The patterns observed in our data challenge the idea that school choice is the primary driver of gender differences in STEM trajectories. Although some single-sex schools offer a narrower range of practical STEM subjects, this cannot account for the substantial gender gaps we observe across the broader student population. In fact, the persistence of these gaps in mixed-sex schools, where male and female students face the same set of subject options, suggests that differences in individual choices rather than school-level constraints play a more important role in explaining gender differences in STEM participation. This implies that the patterns documented in Section 3 are primarily driven by JC subject choices rather than by differences arising from school choice.

Nevertheless, concerns may remain that male and female students sort into schools that differ either in the subjects they offer or in some unobservable characteristics which affect subject choice. To address this directly, we repeat our analysis using more restrictive samples in Table 8. We limit the sample to mixed-sex schools and re-estimate our practical STEM participation models. In addition, we include school fixed effects in these specifications. This approach allows us to examine the gender gap within schools by comparing male and female students who face the same school-level institutional environment and subject

availability. By relying on within-school variation, this specification removes cross-school differences that might otherwise influence subject choice.

Table 8 Practical STEM — Gender gap and Gelbach decomposition Mixed Schools (School Fixed Effects)

	LC	**JC**
Panel A: OLS estimates		
Baseline with School FEs - Male (1)	0.336***	0.625***
	(0.027)	(0.026)
Full Model with School FEs - Male (2)	0.086**	0.605***
	(0.037)	(0.031)
Observations	1,988	1,988
Panel B: Gelbach decomposition of the change in the gender gap		
Total Change: (1)–(2)	0.250***	0.020*
	(0.029)	(0.015)
Contribution to change:		
STEM-Readiness	0.218***	–
	(0.029)	–
Skills/Comparative Advantage	-0.004	0.001
	(0.008)	(0.004)
Preferences/Tastes	0.044**	0.018
	(0.021)	(0.015)
Background Characteristics	-0.008	-0.0002
	(0.005)	(0.004)

Note: Results are estimated on a restricted sample of students attending mixed-sex schools only. All specifications include school fixed effects, which absorb school-level characteristics common to students within the same school, including subject availability and school type. Panel A reports the coefficient on the male indicator from (i) a baseline specification including only gender and school fixed effects and (ii) a full specification including controls. Panel B reports a Gelbach (2016) decomposition of the difference in the Male coefficient between the baseline and full specification, allocating the total change across covariate groups. Blank cells indicate covariate groups not included in the JC specification. Robust standard errors clustered at the school level are reported in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Our results remain qualitatively unchanged across specifications. Even when we limit the sample to mixed-sex schools and include school fixed effects, the estimated gender gaps are similar in magnitude to those in the main analysis. The Gelbach decomposition still highlights the role of prior subject choices in explaining gender differences in LC practical STEM participation. As shown in Table 8, over 64% of the baseline gender gap in LC STEM enrolment (about 87% of the explained component) stems from earlier subject choices at the JC stage.

In contrast, the JC gender gap decomposition shows most of the difference remains unexplained by characteristics observed at age nine. In the "JC" column of Table 8, we see that approximately 96% of the gender gap in JC practical STEM enrolment cannot be explained by early-life background, ability, or preference indicators. Taken together, these results suggest that school-level factors such as subject availability or school selection do not drive the gender gap. When we restrict the sample to mixed-sex schools and exploit within-school variation using school fixed effects, the estimated gaps and decomposition results remain largely unchanged, reinforcing the conclusion that the divergence arises early in students' educational trajectories and is not explained by the observed characteristics we consider.

6 Conclusion

We examine gender disparities in STEM subject enrolment in upper secondary school, a key predictor of STEM participation in higher education. While STEM is often treated as a single category in the literature, it encompasses a diverse set of disciplines, and the gender gaps are considerably larger in practical subjects than in traditional science and maths. We identify the key predictors of enrolment in practical STEM subjects during secondary education using rich data on individuals' abilities, interests, preferences, self-perceived ability, prior experiences, background characteristics, and school-level factors. We find that subject choices made at the lower secondary level, typically around age 13, are the strongest predictor of practical STEM participation at the upper secondary level. These early choices account for 50–60% of the observed gender gap in practical STEM enrolment at upper secondary school.

At the lower secondary level, the rich set of variables observed at age 9 explains only a small share of the gender gap in practical STEM enrolment: over 96% of the gap remains unexplained. This is a notable result given the breadth of the GUI data. In this setting, the results provide little support for several common explanations for gender differences in STEM-related choices: the early practical STEM gap is not primarily explained by measured differences in ability, broad early preferences, or family background. Separate within-school analyses also provide little evidence that school-level sorting accounts for the gap. This suggests that factors beyond early observable characteristics drive divergent curricular choices between boys and girls by early adolescence, a divergence that has materialised by age 13.

Our findings are also robust to a richer and more flexible modelling strategy and to within-school comparisons. Expanding the covariate set in the machine-learning exercise leaves the explained share of the JC gap essentially unchanged, suggesting that the result is not an artefact of our baseline variable selection or linear specification. Likewise, the gender gap remains large in mixed-sex schools and persists when we compare students within the same school using fixed effects. These results indicate that the early divergence in practical STEM subject choice is not driven by differences in school environments, subject availability, or sorting across schools.

The findings of this paper offer clear guidance for policy design. If efforts to reduce gender disparities in practical STEM fields are to succeed, timing is critical. Gender differences in enrolment emerge as soon as subject choice becomes available around age 13, despite similar academic profiles, measured preferences, and background characteristics at age nine. This suggests that interventions should be introduced before subject choices are made, in the years preceding lower secondary school. The evidence points away from standard skill-based and school-constraint explanations and toward mechanisms that are less well captured by conventional survey measures, such as gender norms, stereotypes, and differential social encouragement around practical subjects. Policies should therefore address preferences and attitudes through encouragement, information provision, and efforts to challenge gender stereotypes, rather than initiatives aimed primarily at skill development, academic streaming, or structural changes to school types.

In conclusion, although this study is descriptive rather than causal, it highlights a critical window for intervention and points to concrete policy levers. The persistence of large, unexplained gender gaps in practical STEM participation, even among students with comparable early academic profiles, underscores the need to directly shape perceptions, attitudes, and aspirations before students make subject choices. Secondary school subject selection marks a key juncture in the STEM pipeline, and early, targeted engagement is likely essential to reducing entrenched gender disparities and strengthening the future STEM workforce.

Statements and Declarations

Funding. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Competing interests. The authors have no relevant financial or non-financial interests to disclose.

Ethics approval. Ethics approval was not required for this study.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The data used in this study are Research Microdata Files from the Growing Up in Ireland study, provided by the Central Statistics Office (CSO). These data are not publicly available, but researchers may apply to the CSO for access, subject to its access conditions and confidentiality requirements. Growing Up in Ireland (GUI) is funded by the Government of Ireland. GUI is managed as a partnership between the Department of Children, Disability and Equality (DCDE) and the Central Statistics Office (CSO). The CSO are responsible for the survey element of GUI. Results in this paper are based on analyses of data from Research Microdata Files provided by the CSO. Neither the CSO nor DCDE take any responsibility for the views expressed or the outputs generated from these analyses.

Materials availability. Not applicable.

Code availability. Analysis code is available upon request.

Author contributions. **Dáire Crotty:** Conceptualization, Methodology, Formal analysis and investigation, Writing – original draft preparation. **Ciarán Murphy:** Conceptualization, Methodology, Formal analysis and investigation, Writing – original draft preparation.

Appendix A Additional Results and Information

A.1 Summary Statistics and Information



Figure 1 Subject Availability by School Gender Composition - Upper and Lower Secondary

Table 9 Summary Statistics of School Characteristics

	Total (%)	Pre (%)	Post (%)	Weighted - Post (%)
School Gender				
Single Sex - Boys	16.01	22.87	23.34	23.23
Single Sex - Girls	20.65	24.98	25.58	25.64
Mixed Sex	63.34	52.15	51.08	51.13

Note: Distribution of school gender composition in the sample and in national statistics. "Total" refers to national figures for Irish secondary schools. "Pre" refers to the full GUI sample; "Post" refers to the sample after restrictions; "Weighted-Post" applies survey weights to the restricted sample. Total figures are taken from national statistics for the school years 2015/2016 and 2016/2017. For reference see secondary school information via https://data.cso.ie/

Table 10 Observations Lost by Sample Restriction

Variable	*Sample*	*Observations dropped*	*Sample left*
Wave 1			
PCG being the Mother	6,039	71	5,968
Books in home	5,968	2	5,966
PCG Foreign Born	5,966	4	5,962
Like Maths	5,962	21	5,941
Drumcondra Numerical Score	5,941	58	5,883
Drumcondra Reading Score	5,883	69	5,814
Wave 2			
Academic Self-Concept	5,814	69	5,745
Conscientiousness	5,745	6	5,739
Practical favourite subject	5,739	182	5,577
Own expectations	5,577	83	5,494
Maths Difficult	5,494	6	5,488
English Difficult	5,488	3	5,485
Wave 3			
Doing English	5,485	41	5,444
Doing Higher English	5,444	2	5,442
Doing Maths	5,442	6	5,436
Transition Year	5,436	65	5,371
Number of Higher-Level Subjects	5,371	13	5,358
English JC Grade	5,358	39	5,319
Maths JC Grade	5,319	2	5,317
Principal - School Gender	5,317	1,385	3,932
Principal - School Type	3,932	37	3,895
Principal - DEIS Status	3,895	7	3,888

Note: The table documents the sequential sample restrictions applied to construct the dataset. The initial balanced panel consists of 6,039 students participating in the first three waves of GUI. Each row shows the number of observations dropped due to missing data for a given variable and the remaining sample size after the restriction. A large number of principals did not answer the questionnaire relating to school characteristics and subject offerings. Hence, this is where we see the largest loss of observations. PCG = Primary Caregiver.

Table 11 Distribution of students per school

Students per school	*Number of schools*	*Percent*
1	38	9.11
2	32	7.67
3	32	7.67
4	27	6.47
5	28	6.71
6	23	5.52
7	27	6.47
8	18	4.32
9	20	4.80
10	23	5.52
11	17	4.08
12	19	4.56
13	12	2.88
14	15	3.60
15	16	3.84
16	11	2.64
17	15	3.60
18	2	0.48
19	8	1.92
20	4	0.96
21	1	0.24
22	4	0.96
23	3	0.72
24	4	0.96
25	2	0.48
26	2	0.48
27	1	0.24
28	1	0.24
29	2	0.48
30	2	0.48
31	2	0.48
34	2	0.48
37	1	0.24
41	1	0.24
42	2	0.48
Total	417	100.00

A.2 Comparison of Enrolment Rates

Table 12 Subject Enrolment Rates for Practical Subjects - LC & JC

Subject	Gender	2015	2016	2017	2018	Total (Average) %	Pre %	Post %	Weighted - Post %
LC - Total Enrolment									
Total		55,006	55,684	55,731	54,396	-	-	-	-
	Female	27,815	27,790	27,886	27,576	-	-	-	-
	Male	27,192	27,894	27,845	26,820	-	-	-	-
LC - Practical STEM									
Construction Studies		8,269	8,553	8,750	8,248	15.32	11.14	10.23	12.28
	Female	676	763	859	902	1.45	1.06	0.66	0.90
	Male	7,593	7,790	7,891	7,346	13.87	10.08	9.57	11.38
Agricultural Science		7,672	7,893	7,660	7,778	14.04	11.33	10.70	11.90
	Female	2,920	3,048	3,093	3,152	5.53	5.19	4.65	4.79
	Male	4,752	4,845	4,567	4,626	8.51	6.14	6.05	7.11
DCG		5,362	5,523	5,575	5,393	9.90	9.62	9.43	9.72
	Female	649	661	729	760	1.27	1.47	1.61	1.80
	Male	4,713	4,862	4,846	4,633	8.63	8.15	7.82	7.92
Engineering		5,376	5,379	5,275	5,254	9.64	6.68	6.18	7.64
	Female	324	298	315	355	0.59	0.40	0.39	0.39
	Male	5,052	5,081	4,960	4,899	9.05	6.28	5.79	7.25
Technology		1,328	1,415	1,527	1,534	2.63	2.85	2.96	3.38
	Female	233	231	271	247	0.40	0.59	0.52	0.54
	Male	1,095	1,184	1,256	1,287	2.23	2.26	2.44	2.84
	Gender	2012	2013	2014	2015				
JC - Total Enrolment									
Total		58,798	59,822	60,327	59,522	-	-	-	-
	Female	28,891	29,269	29,277	29,213	-	-	-	-
	Male	29,907	30,553	31,050	30,309	-	-	-	-
JC - Practical STEM									
Material Technology		15,775	16,163	16,464	16,145	27.08	21.44	20.07	24.01
	Female	1,964	2,139	2,145	2,500	3.67	2.84	2.18	2.80
	Male	13,811	14,024	14,319	13,645	23.41	18.60	17.89	21.21
Technical Graphics		12,004	12,564	11,629	11,701	20.39	20.03	19.50	19.93
	Female	1,705	1,805	1,758	1,746	2.94	2.97	3.04	3.01
	Male	10,299	10,759	9,871	9,955	17.45	17.06	16.46	16.92
Metalwork		7,739	7,843	7,880	7,984	13.19	9.21	8.34	10.18
	Female	717	623	742	797	1.21	0.86	0.80	0.90
	Male	7,022	7,220	7,138	7,187	11.98	8.35	7.54	9.28
Technology		3,026	2,957	3,223	3,258	5.23	5.22	5.76	6.58
	Female	618	525	531	612	0.96	1.02	0.88	1.01
	Male	2,408	2,432	2,692	2,646	4.27	4.20	4.88	5.57

Note: 'Total' column is the national average. 'Pre' column is the unweighted sample before data restrictions. 'Post' column is the unweighted sample after data restrictions. 'Weighted - Post' column is the weighted sample after data restrictions. National figures are from the State Examinations Commission (SEC) https://www.examinations.ie/statistics/

A.3 Full list of variables

Table 13: Summary of Variables

Variable	Group	Age	LC/JC	Description	Mean
Main Analysis - Variables					
Gender	Gender	9	Both	=1 if student is male	0.49
Studied Foreign Language for JC	STEM-readiness	17/18	LC	=1 if student took JC French, German or Spanish	0.90
Studied Art for JC	STEM-readiness	17/18	LC	=1 if student took JC Art	0.33
Studied Music for JC	STEM-readiness	17/18	LC	=1 if student took JC Music	0.20
Studied Home Economics for JC	STEM-readiness	17/18	LC	=1 if student took JC Home Economics	0.33
Studied Material Technology for JC	STEM-readiness	17/18	LC	=1 if student took JC Material Technology	0.24
Studied Metalwork for JC	STEM-readiness	17/18	LC	=1 if student took JC Metalwork	0.10
Studied Technical Graphics for JC	STEM-readiness	17/18	LC	=1 if student took JC TG	0.20
Studied Business Studies for JC	STEM-readiness	17/18	LC	=1 if student took JC Business Studies	0.59
Studied Technology for JC	STEM-readiness	17/18	LC	=1 if student took JC Technology	0.07
Studied Religion for JC	STEM-readiness	17/18	LC	=1 if student took JC Religion	0.50
Studied Higher-Level English for JC	STEM-readiness	17/18	LC	=1 if student took JC English at Higher Level	0.83
Studied Higher-Level Maths for JC	STEM-readiness	17/18	LC	=1 if student took JC Maths at Higher Level	0.65
Did Transition Year	STEM-readiness	17/18	LC	=1 if student took Transition Year	0.71
Drumcondra Reading Score	Skills/comp. adv.	9	Both	Standardised reading test	0.12
Drumcondra Numerical Score	Skills/comp. adv.	9	Both	Standardised math/numerical	-0.65
Academic Self-Concept	Skills/comp. adv.	13	LC	Academic self-concept scale	12.10
Conscientiousness Score	Skills/comp. adv.	13	LC	Conscientiousness measure	4.23
Strengths & Difficulties Questionnaire (SDQ) Score	Skills/comp. adv.	9	Both	Non-cognitive measure	6.58
Number of Higher-Level Subjects taken for JC	Skills/comp. adv.	17/18	LC	Number of JC subjects at higher level, 0–12	8.21
JC English Grade	Skills/comp. adv.	17/18	LC	Letter grade converted	9.87
JC Maths Grade	Skills/comp. adv.	17/18	LC	Letter grade converted	9.30
School Gender 1	School/Peer Effects	17/18	LC	=1 if Male-only school	0.23

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Table 13 – continued from previous page

Variable	Group	Age	LC/JC	Description	Mean
School Gender 2	School/Peer Effects	17/18	LC	=1 if Female-only school	0.26
School Gender 3	School/Peer Effects	17/18	LC	=1 if Mixed-sex school	0.51
School Type 1	School/Peer Effects	17/18	LC	=1 if Fee-paying school	0.11
School Type 2	School/Peer Effects	17/18	LC	=1 if Non-fee-paying secondary	0.49
School Type 3	School/Peer Effects	17/18	LC	=1 if Vocational school	0.12
School Type 4	School/Peer Effects	17/18	LC	=1 if Community/comprehensive	0.27
Attended DEIS School	School/Peer Effects	17/18	LC	=1 if DEIS school	0.15
French Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC French	0.99
German Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC German	0.67
Spanish Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Spanish	0.42
Art Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Art	0.95
Applied Maths Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Applied Maths	0.45
Physics Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Physics	0.92
Chemistry Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Chemistry	0.94
Agricultural Science Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Agricultural Science	0.66
Biology Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Biology	0.99
Engineering Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Engineering	0.47
Construction Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Construction	0.65
Accounting Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Accounting	0.88
Business Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Business	0.97
Economics Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Economics	0.61
Technology Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Technology	0.22
Music Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Music	0.88
Home Economics Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Home Economics	0.77
Project Maths Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Project Maths	0.73
Religious Studies Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Religious Studies	0.51
Link Modules Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC Link Modules	0.63
DCG Offered in School	School/Peer Effects	17/18	LC	=1 if school offers LC DCG	0.71
Other Languages Offered in School	School/Peer Effects	17/18	LC	=1 if school offers other LC languages	0.08

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Table 13 – continued from previous page

Variable	Group	Age	LC/JC	Description	Mean
YP Likes Maths	Preferences/Tastes	9	Both	1=Always, 2=Sometimes, 3=Never	1.61
YP Likes Reading	Preferences/Tastes	9	Both	1=Always, 2=Sometimes, 3=Never	1.46
What YP thinks about School	Preferences/Tastes	9	Both	1=Always like it, 2=Sometimes like it, 3=Never like it	1.81
YP Preferred adult occupation 1	Preferences/Tastes	9	Both	Teaching	0.13
YP Preferred adult occupation 2	Preferences/Tastes	9	Both	Health & care	0.11
YP Preferred adult occupation 3	Preferences/Tastes	9	Both	Professional STEM	0.04
YP Preferred adult occupation 4	Preferences/Tastes	9	Both	Technical STEM	0.01
YP Preferred adult occupation 5	Preferences/Tastes	9	Both	Trades/construction/mechanic	0.04
YP Preferred adult occupation 6	Preferences/Tastes	9	Both	Transport/agriculture/production	0.05
YP Preferred adult occupation 7	Preferences/Tastes	9	Both	Arts/media	0.38
YP Preferred adult occupation 8	Preferences/Tastes	9	Both	Sport	0.01
YP Preferred adult occupation 9	Preferences/Tastes	9	Both	Protective services	0.05
YP Preferred adult occupation 10	Preferences/Tastes	9	Both	Business/law/management	0.04
YP Preferred adult occupation 11	Preferences/Tastes	9	Both	Clerical/service admin	0.01
YP Preferred adult occupation 12	Preferences/Tastes	9	Both	Don't know/uncodeable	0.14
YP Practical Favourite Subject	Preferences/Tastes	13	LC	=1 if favourite JC subject is a practical-based subject	0.18
YP Own Expectations regarding their education	Preferences/Tastes	13	LC	Educational expectations (1–6)	3.29
Maths Difficult	Preferences/Tastes	13	LC	1=Difficult, 2=OK, 3=Not difficult	2.27
English Difficult	Preferences/Tastes	13	LC	1=Difficult, 2=OK, 3=Not difficult	2.53
Disability status of YP	Background	9	Both	=1 if child has a disability	0.10
Immigrant Status of PCG	Background	9	Both	=1 if PCG was born outside Ireland	0.15
Number of Books in home	Background	9	Both	Number of books (bands)	4.25
PCG is university level educated	Background	9	Both	=1 if mother has a university degree	0.18
Robustness Testing/Machine Learning - Additional Variables					
Teacher assessment of YP: reading	Skills/comp. adv.	9	Both	Teacher assessment of child's reading. 1=Below average, 2=average, 3=above average	2.29

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Table 13 – continued from previous page

Variable	Group	Age	LC/JC	Description	Mean
Teacher assessment of YP: writing	Skills/comp. adv.	9	Both	Teacher assessment of child's writing. 1=Below average, 2=average, 3=above average	2.18
Teacher assessment of YP: comprehension	Skills/comp. adv.	9	Both	Teacher assessment of child's comprehension. 1=Below average, 2=average, 3=above average	2.27
Teacher assessment of YP: maths	Skills/comp. adv.	9	Both	Teacher assessment of child's maths. 1=Below average, 2=average, 3=above average	2.23
Teacher assessment of YP: creativity	Skills/comp. adv.	9	Both	Teacher assessment of child's creativity. 1=Below average, 2=average, 3=above average	2.28
Teacher assessment of YP: oral communication	Skills/comp. adv.	9	Both	Teacher assessment of child's oral communication. 1=Below average, 2=average, 3=above average	2.31
Teacher assessment of YP: problem solving	Skills/comp. adv.	9	Both	Teacher assessment of child's problem-solving. 1=Below average, 2=average, 3=above average	2.13
PCG perception of YP: above-average math	Skills/comp. adv.	9	Both	=1 if PCG rates child's maths ability as above average or excellent (relative to other children of same age)	0.55
PCG perception of YP: above average reading	Skills/comp. adv.	9	Both	=1 if PCG rates child's reading ability as above average or excellent (relative to other children of same age)	0.61
BAS Score	Skills/comp. adv.	13	LC	BAS Score measure	115.19
SDQ Score - Total from PCG questions	Skills/comp. adv.	13	LC	Non-cognitive measure	7.54
Parent/guardian expectations of YP	Preferences/Tastes	9	Both	How far PCG expects their child will go in education 1=Junior Certificate or equivalent, 2=Leaving Certificate or equivalent, 3=An apprenticeship or trade, 4=Diploma/Certificate, 5=Degree, 6=Postgraduate/higher degree	4.74
Under 1 hour TV per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour watching television each day	0.24
Under 1 hour reading per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour reading each day	0.67
Under 1 hour computer per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour on computer each day	0.86
Under 1 hour gaming per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour on video games each day	0.79
Under 1 hour homework per day	Preferences/Tastes	9	Both	=1 if child spends <1 hour on homework each day	0.74
Computer in bedroom	Preferences/Tastes	9	Both	=1 if child has computer in room	0.07
Games console in bedroom	Preferences/Tastes	9	Both	=1 if child has games console in room	0.32

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Table 13 – continued from previous page

Variable	Group	Age	LC/JC	Description	Mean
Sports/fitness club	Preferences/Tastes	9	Both	=1 if during an average week, child participates in sports/fitness club	0.78
Cultural activities	Preferences/Tastes	9	Both	=1 if during an average week, child participates in cultural activities	0.50
Youth club member	Preferences/Tastes	9	Both	=1 if during an average week, child participates in youth club	0.06
Scouts member	Preferences/Tastes	9	Both	=1 if during an average week, child participates in scouts/guides etc.	0.15
Homework club member	Preferences/Tastes	9	Both	=1 if during an average week, child participates in homework club	0.06
Own bedroom	Preferences/Tastes	9	Both	=1 if child has their own bedroom	0.67
Self: doing well in schoolwork	Preferences/Tastes	9	Both	How does child think they're doing in schoolwork? 1=Well, 2=Average/ok, 3=Poorly	1.37
Like Irish	Preferences/Tastes	9	Both	1=Child always likes Irish, 2=Sometimes, 3=Never	2.04
Look forward to school	Preferences/Tastes	9	Both	1=Child always looks forward to school, 2=Sometimes, 3=Never	1.86
Like teacher	Preferences/Tastes	9	Both	1=Child always likes teacher, 2=Sometimes, 3=Never	1.52
Top 3 activities: friends	Preferences/Tastes	9	Both	=1 if child rates hanging out with friends as one of top 3 best things to do in spare time	0.65
Top 3 activities: chat online	Preferences/Tastes	9	Both	=1 if child rates chatting to friends on phone/computer as one of top 3 best things to do in spare time	0.09
Top 3 activities: sport	Preferences/Tastes	9	Both	=1 if child rates playing sport as one of top 3 best things to do in spare time	0.62
Top 3 activities: TV	Preferences/Tastes	9	Both	=1 if child rates watching television as one of top 3 best things to do in spare time	0.30
Top 3 activities: gaming	Preferences/Tastes	9	Both	=1 if child rates playing video games as one of top 3 best things to do in spare time	0.18
Top 3 activities: reading	Preferences/Tastes	9	Both	=1 if child rates reading as one of top 3 best things to do in spare time	0.25
Top 3 activities: outdoors	Preferences/Tastes	9	Both	=1 if child rates playing games outside as one of top 3 best things to do in spare time	0.44

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Table 13 – continued from previous page

Variable	Group	Age	LC/JC	Description	Mean
Top 3 activities: listening to music	Preferences/Tastes	9	Both	=1 if child rates listening to music as one of top 3 best things to do in spare time	0.19
Top 3 activities: family	Preferences/Tastes	9	Both	=1 if child rates talking to family as one of top 3 best things to do in spare time	0.22
Read daily	Preferences/Tastes	9	Both	=1 if child reads for fun daily	0.32
Least Favourite Subject	Preferences/Tastes	13	LC	=1 if YP report maths or science as their least favourite subject in school	0.20
Maths Interesting	Preferences/Tastes	13	LC	1=Child thinks Maths is interesting, 2 =Ok, 3=Not Interesting	1.85
English Interesting	Preferences/Tastes	13	LC	1=Child thinks English is Interesting, 2 =Ok, 3=Not Interesting	1.67
Single-sex school	School/Peer Effects	9	Both	=1 if primary school is single-sex	0.46
Transition Year Offered in School	School/Peer Effects	17/18	LC	=1 if school offers Transition Year	0.98
Parent/Guardian Disability status	Background	9	Both	=1 if primary caregiver has a disability	0.12
Immigrant Status of YP	Background	9	Both	=1 if child is immigrant	0.11
Dyslexic Status of YP	Background	9	Both	=1 if child has dyslexia	0.03
ADHD Status of YP	Background	9	Both	=1 if child has ADHD	0.00
Autism Status of YP	Background	9	Both	=1 if child has autism	0.01
YP has Speech/language difficulty	Background	9	Both	=1 if child has speech/language difficulties	0.01
Dyspraxia Status of YP	Background	9	Both	=1 if child has dyspraxia	0.01
Slow progress Status of YP	Background	9	Both	=1 if child has slow progress	0.02
Household type 1	Background	9	Both	=1 if Single parent plus 1–2 children	0.09
Household type 2	Background	9	Both	=1 if Single parent plus 3+ children	0.05
Household type 3	Background	9	Both	=1 if Couple plus 1–2 children	0.36
Household type 4	Background	9	Both	=1 if Couple plus 3+ children	0.49
5 or more in household	Background	9	Both	=1 if >4 people in household	0.58
YP is Regularly unwell	Background	9	Both	=1 if child's health rated worse than 'Very healthy, no problems'	0.26
YP has Sight issue	Background	9	Both	=1 if child has/has had sight issue requiring correction	0.16
YP has Hearing issue	Background	9	Both	=1 if child has/has had hearing issue requiring correction	0.08

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Table 13 – continued from previous page

Variable	Group	Age	LC/JC	Description	Mean
YP has Speech issue	Background	9	Both	=1 if PCG worries about how child talks and makes speech sounds	0.07
PCG basic calculations	Background	9	Both	=1 if PCG can tell if they have the correct change when buying things with €5 or €10	0.98
Parents meet teacher regularly	Background	9	Both	=1 PCG or spouse attended formal meeting with teacher in last academic year	0.97
Under 3 absent days	Background	9	Both	=1 if child was absent for <3 days in last academic year	0.73
Parents always help with homework	Background	9	Both	=1 if PCG or spouse always/nearly always helps child with homework	0.49

Note: The column labelled "LC or JC" indicates whether a variable is included in the lower secondary (JC) or upper secondary (LC) analysis. For the lower secondary analysis, explanatory variables are restricted to measures collected in Wave 1 (age 9), meaning that only age-9 variables are common to both analyses. Variables collected in Wave 2 or Wave 3 (ages 13 and 17/18, respectively) are used exclusively in the upper secondary analysis.

A.4 eXtreme Gradient Boosting

Students' decisions are influenced by a wide range of personal and school-level characteristics that interact in non-obvious, often nonlinear ways. Standard regression techniques can be poorly suited to this context: they require prior specification of which variables matter and how they relate to the outcome, assumptions that are difficult to justify with such rich data. Moreover, including many predictors in a linear model risks collinearity, overfitting, and reduced interpretability. To navigate these challenges, we adopt the machine learning method eXtreme Gradient Boosting (XGBoost) (Chen and Guestrin 2016), which accommodates complex interactions and nonlinearities without predefining relationships between variables.

XGBoost constructs a predictive model by building a sequence of decision trees, each improving on the previous one. The final prediction is a weighted sum of individual tree contributions. It identifies patterns in the training data (70% of our sample) and uses them to forecast STEM subject choices.[11] Each tree divides the data based on values of predictor variables, grouping students with similar likelihoods of studying STEM. While individual decision trees can overfit the training data, XGBoost avoids this by combining many shallow trees, each correcting the previous ones' mistakes. This process allows it to capture subtle relationships while avoiding the pitfalls of more flexible but less disciplined models.

We select hyperparameters to balance flexibility with generalisability. In particular, *max_depth* controls how deep each decision tree may grow. We tune this hyperparameter by cross-validation; we search over values $\{3, 4, 5, 6, 7\}$ using 5-fold cross-validation on the training data and select the value that maximises the area under the ROC curve (AUC). All remaining hyperparameters follow standard applied practice to prevent overfitting: the learning rate is $\eta = 0.05$; the number of boosting rounds is fixed at $nrounds = 100$; each tree uses a random 80% subsample of the training observations ($subsample = 0.8$) and a random 80% subset of predictors ($colsample_bytree = 0.8$); and $\gamma = 0$, $min_child_weight = 1$. Before fitting, we handle missing values by median imputation for continuous variables and modal-category imputation for categorical variables.[12]

Table 14 Predictive Performance of XGBoost Models

Model	NIR	Test (Acc)	Test (AUC)	Sensitivity	Specificity	Precision	F1 Score	Brier
JC	0.65	78.23	0.86	64.32	85.79	71.11	67.55	0.144
LC	0.71	77.43	0.80	43.64	91.38	67.61	53.04	0.158

Note: This table reports out-of-sample predictive performance for the XGBoost models used in the SHAP decomposition analysis. NIR denotes the no-information rate. Sensitivity measures the share of true STEM participants correctly classified, while specificity measures the share of non-STEM students correctly classified. The JC model exhibits a higher AUC, F1 score, and lower Brier score, indicating more balanced predictive performance overall, whereas the LC model achieves higher specificity but lower sensitivity.

Both models achieve similar levels of predictive accuracy (approximately 77–78%) and outperform the no-information rate, indicating meaningful predictive performance. However, the JC STEM model demonstrates a higher AUC (0.86 versus 0.80) and substantially greater sensitivity, suggesting it is more effective at identifying STEM participants. The LC STEM model achieves higher specificity, indicating stronger performance in correctly classifying non-STEM outcomes. Overall, the JC model achieves a higher F1 score and lower Brier score, indicating a more balanced and better-calibrated model. In contrast, the LC model appears more conservative, prioritising correct classification of the majority class.

[11] We conduct sensitivity tests using an alternative 80/20 split, confirming the stability and consistency of our findings.

[12] Of the 139 predictors, 115 have no missing values, and a further 14 have fewer than 1% of observations imputed. The following 10 variables have higher imputation rates: teacher assessments of reading (3.48%), writing (3.53%), comprehension (3.50%), maths (3.63%), creativity (3.58%), oral communication (3.79%), and problem solving (4.05%); look forward to school (1.58%); like teacher (1.61%); and BAS score (1.61%).

A.5 SHAP Decompositions

To fully capitalise on the flexible, nonlinear nature of the XGBoost model, we use SHAP values for decomposition analysis. SHAP values quantify each predictor's contribution to the predicted probability of STEM enrolment and satisfy fairness criteria derived from cooperative game theory.

We perform the decomposition on the held-out test set (the 30% of observations not used in training). For each predictor x_i, we compute its weighted mean SHAP value separately for female and male students. The contribution of variable x_i to the predicted gender gap is:

$$\bar{\phi}_i^{\mathrm{M}} - \bar{\phi}_i^{\mathrm{F}}$$

where $\bar{\phi}_i^{\mathrm{M}}$ and $\bar{\phi}_i^{\mathrm{F}}$ are the weighted mean SHAP values for male and female students, respectively. Summing these contributions across all predictors recovers the total predicted gender gap. We then aggregate individual predictors into the variable groups defined in Appendix Table 13 (STEM-readiness, skills/comparative advantage, school factors/peer effects, preferences/tastes, and background characteristics). The SHAP value for the gender indicator captures the residual direct contribution of gender not explained by the other included variables.

Traditional decomposition methods typically dissect the observed enrolment gap. In contrast, our SHAP-based decomposition analyses the gap predicted by the XGBoost model, fully utilising all available predictors and their interactions without restrictive assumptions such as linearity or preselected variables. While traditional decomposition methods rely heavily on linearity assumptions, SHAP decomposition does not. However, the validity of the SHAP decomposition hinges on the accuracy of the underlying model predictions. If the model systematically mispredicts enrolment probabilities differently by gender, the decomposed predicted gap may differ from the true observed gap. Therefore, our results section clearly distinguishes between the observed and predicted gaps, enabling a transparent assessment of potential discrepancies.

A.6 Regression Results - LC & JC

Table 15: Determinants of LC Practical STEM Participation

	Coef	SE
Male	0.071**	(0.034)
Group 1 – STEM-Readiness		
Studied Foreign Language for JC	-0.028	(0.040)
Studied Art for JC	-0.044*	(0.024)
Studied Music for JC	-0.039	(0.024)
Studied Home Economics for JC	-0.030	(0.025)
Studied Material Technology for JC	0.110***	(0.034)
Studied Metalwork for JC	0.141***	(0.041)
Studied Technical Graphics for JC	0.230***	(0.031)
Studied Technology for JC	0.198***	(0.052)
Studied Business Studies for JC	-0.036	(0.023)
Studied Religion for JC	-0.005	(0.021)
Studied Higher-Level English for JC	-0.005	(0.045)
Studied Higher-Level Maths for JC	-0.059*	(0.032)
Did Transition Year	0.040*	(0.021)
Group 2 – Skills/Comparative Advantage		
Drumcondra Numerical Score	-0.012	(0.012)
Drumcondra Reading Score	-0.031**	(0.013)
Academic Self-Concept Score	-0.000	(0.003)
Conscientiousness Score	-0.007	(0.004)
SDQ Score	-0.002	(0.002)
Number of Higher Level Subjects Taken for JC	0.013**	(0.006)
JC English Grade	-0.004	(0.010)
JC Maths Grade	-0.004	(0.009)
Group 3 – School Factors/Peer Effects		
School Gender (ref: Girls-only)		
Boys-only	0.026	(0.057)
Mixed-sex	0.016	(0.042)
School Type (ref: Vocational)		
Fee-paying	-0.004	(0.050)
Non-fee-paying	-0.025	(0.040)
Community/comprehensive	-0.031	(0.036)
DEIS status of school	0.003	(0.026)
French Offered in School	-0.070	(0.095)
German Offered in School	0.016	(0.023)
Spanish Offered in School	-0.005	(0.021)
Art Offered in School	-0.009	(0.042)
Applied Maths Offered in School	-0.015	(0.021)
Physics Offered in School	0.041	(0.045)
Chemistry Offered in School	-0.009	(0.073)
Agricultural Science Offered in School	0.086***	(0.021)
Biology Offered in School	0.211***	(0.074)

Continued on next page

Table 15 continued

	Coef	SE
Engineering Offered in School	0.023	(0.036)
Construction Studies Offered in School	-0.003	(0.036)
Accounting Offered in School	0.012	(0.037)
Business Offered in School	-0.046	(0.058)
Economics Offered in School	-0.012	(0.024)
Technology Offered in School	0.013	(0.026)
Music Offered in School	-0.003	(0.033)
Home Economics Offered in School	0.063	(0.039)
Project Maths Offered in School	0.012	(0.023)
Religious Studies Offered in School	-0.009	(0.019)
Link Modules Offered in School	-0.003	(0.020)
DCG Offered in School	0.014	(0.030)
Other Languages Offered in School	-0.051	(0.041)
Group 4 – Preferences/Tastes		
Like Maths (ref: Always)		
Like Maths (Sometimes)	0.022	(0.019)
Like Maths (Never)	-0.080**	(0.031)
Like Reading (ref: Always)		
Like Reading (Sometimes)	0.016	(0.018)
Like Reading (Never)	-0.026	(0.047)
Practical Favourite Subject	0.146***	(0.034)
Own Expectations of educational attainment (ref: JC)		
Own Expectations (LC)	-0.052	(0.060)
Own Expectations (Higher Certificate)	-0.025	(0.056)
Own Expectations (Degree or higher degree)	-0.062	(0.057)
Maths Difficult (ref: Difficult)		
Maths Difficult (Ok)	0.032	(0.028)
Maths Difficult (Not Difficult)	0.042	(0.032)
English Difficult (ref: Difficult)		
English Difficult (Ok)	0.021	(0.049)
English Difficult (Not Difficult)	-0.002	(0.048)
Like School (ref: Always)		
Like School (Sometimes)	-0.014	(0.022)
Like School (Never)	0.035	(0.042)
Occupation (ref: Teaching)		
YP Preferred adult occupation – Healthcare	-0.005	(0.033)
YP Preferred adult occupation – Professional STEM	0.084	(0.064)
YP Preferred adult occupation – Technical STEM	-0.112	(0.177)
YP Preferred adult occupation – Trades/construction/mechanics	0.085	(0.052)
YP Preferred adult occupation – Transport/agricultural/production	0.102**	(0.052)
YP Preferred adult occupation – Arts/Media	0.012	(0.028)
YP Preferred adult occupation – Sports	-0.147***	(0.040)
YP Preferred adult occupation – Protective Services	-0.003	(0.046)
YP Preferred adult occupation – Business/law/management	-0.038	(0.052)
YP Preferred adult occupation – Clerical/service admin	0.090	(0.129)

Continued on next page

Table 15 continued

	Coef	SE
YP Preferred adult occupation – Other/unknown	0.002	(0.030)
Group 5 – Background Characteristics		
YP has a Disability	0.005	(0.030)
PCG is born outside of Ireland	-0.072***	(0.026)
PCG is University level Educated	0.010	(0.023)
Books in Home (ref: None)		
Number of Books in Home (¡10)	0.332	(0.290)
Number of Books in Home (10-20)	0.333	(0.288)
Number of Books in Home (21-30)	0.336	(0.289)
Number of Books in Home (¿30)	0.342	(0.289)
R-Squared	0.30	
Observations	3,888	

Note: Linear probability model estimates of participation in practical STEM subjects at Leaving Certificate level. The dependent variable is an indicator equal to one if the student takes at least one practical STEM subject and zero otherwise. Robust standard errors clustered by school in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 16: Determinants of JC Practical STEM Participation

	Coef	SE
Male	0.570***	(0.032)
Group 1 – Skills/Comparative Advantage		
Drumcondra Numerical Score	-0.010	(0.012)
Drumcondra Reading Score	-0.046***	(0.014)
Group 2 – Preferences/Tastes		
Like Maths (ref: Always)		
Like Maths (Sometimes)	-0.000	(0.019)
Like Maths (Never)	-0.047	(0.030)
Like Reading (ref: Always)		
Like Reading (Sometimes)	-0.014	(0.019)
Like Reading (Never)	-0.066	(0.046)
Like School (ref: Always)		
Like School (Sometimes)	0.019	(0.020)
Like School (Never)	0.068*	(0.038)
Occupation (ref: Teaching)		
YP Preferred adult occupation – Healthcare	-0.015	(0.031)
YP Preferred adult occupation – Professional STEM	0.072	(0.058)
YP Preferred adult occupation – Technical STEM	0.233*	(0.138)
YP Preferred adult occupation – Trades/construction/mechanics	0.101**	(0.047)
YP Preferred adult occupation – Transport/agricultural/production	0.050	(0.038)
YP Preferred adult occupation – Arts/Media	0.028	(0.026)
YP Preferred adult occupation – Sports	-0.083	(0.081)
YP Preferred adult occupation – Protective Services	0.006	(0.045)

Continued on next page

Table 16 continued

	Coef	SE
YP Preferred adult occupation – Business/law/management	0.030	(0.061)
YP Preferred adult occupation – Clerical/service admin	-0.086	(0.079)
YP Preferred adult occupation – Other/unknown	0.045	(0.033)
Group 3 – Background Characteristics		
YP has a Disability	-0.036	(0.030)
PCG is born outside of Ireland	0.038	(0.024)
PCG is University level Educated	-0.079***	(0.029)
Books in Home (ref: None)		
Number of Books in Home (¡10)	-0.050	(0.127)
Number of Books in Home (10-20)	-0.050	(0.125)
Number of Books in Home (21-30)	-0.042	(0.125)
Number of Books in Home (¿30)	-0.056	(0.123)
R-Squared	0.37	
Observations	3,888	

Note: Linear probability model estimates of participation in practical STEM subjects at Junior Certificate level. The dependent variable is an indicator equal to one if the student takes at least one practical STEM subject and zero otherwise. Robust standard errors clustered by school in parentheses. Statistical significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.7 Upper and Lower Secondary Gender Gap - By Individual Subject

Appendix Table 17 extends the analysis by examining the gender gap separately across individual practical STEM subjects at the Leaving Certificate level: Technology, Construction Studies, Engineering, Design and Communication Graphics (DCG), and Agricultural Science. In the baseline specification, substantial gender differences are observed across all subjects. The largest disparities appear in Construction Studies and Engineering, where the male coefficients are 20.40 and 13.47 percentage points, respectively, while smaller but still sizeable gaps exist in Technology (4.52), DCG (12.00), and Agricultural Science (4.85). Introducing the full set of covariates substantially reduces the estimated gender differences in several subjects. In Technology and DCG, the gender gap is almost entirely eliminated and becomes statistically insignificant. In Construction Studies and Engineering, the gender disparity remains statistically significant but is considerably smaller in magnitude. Agricultural Science also shows a large reduction in the gender coefficient once observable characteristics are included.

As in the main analysis, Panel B reports the decomposition of these changes. Across subjects, the dominant contributor to the reduction in the gender coefficient is STEM-readiness, captured primarily by Junior Cycle subject choices and related indicators. For Construction Studies, STEM-readiness accounts for nearly all of the reduction in the gender gap, explaining 11.94 percentage points of the total change. Similar patterns emerge in Technology and DCG, where earlier subject choices explain the majority of the observed reduction in the gender coefficient. In Engineering, STEM-readiness accounts for the largest share of the change, though additional mechanisms also play a role. Even in Agricultural Science, where the overall gender gap is smaller, earlier subject preparation explains most of the reduction observed when controls are introduced.

Other mechanisms play a more limited and heterogeneous role across subjects. Measures of skills contribute very little to explaining the gender gap across the subjects considered. The school factors/peer effects group enters with a negative sign in Construction Studies and Engineering, indicating it widens rather than narrows the explained portion of the gap. Preference-related variables also explain a non-trivial share of the change in several subjects, particularly Construction Studies, Engineering, and DCG.

However, their contribution is smaller than that of STEM-readiness. Background characteristics explain little variation across subjects.

Taken together, these subject-level results reinforce the pipeline interpretation observed in the main analysis. Much of the gender disparity in practical STEM participation at the end of secondary school can be traced to earlier differences in curricular pathways established during lower secondary education. Subject choices made at the Junior Cycle stage appear central to shaping access to practical STEM subjects later in the educational trajectory.

Table 17 LC Practical STEM — Gender gap and Gelbach decomposition - By Individual Subject

	Technology	Construction	Engineering	DCG	Ag Science
Panel A: OLS estimates					
Baseline - Male (1)	0.045***	0.204***	0.135***	0.120***	0.048***
	(0.010)	(0.017)	(0.014)	(0.015)	(0.016)
Full Model - Male (2)	0.002	0.084***	0.081***	-0.010	0.008
	(0.021)	(0.022)	(0.017)	(0.021)	(0.024)
Observations	3,888	3,888	3,888	3,888	3,888
Panel B: Gelbach decomposition of the change in the gender gap					
Total Change: (1)–(2)	0.043***	0.120***	0.054***	0.130***	0.041
	(0.016)	(0.022)	(0.020)	(0.021)	(0.025)
Contribution to change:					
STEM-Readiness	0.029***	0.119***	0.064***	0.085***	0.033**
	(0.010)	(0.017)	(0.015)	(0.010)	(0.016)
Skills/Comparative Advantage	0.002	-0.002	0.0008	-0.0000	0.0006
	(0.002)	(0.003)	(0.002)	(0.003)	(0.004)
School Factors/Peer Effects	0.003	-0.030**	-0.036***	0.012	-0.0003
	(0.009)	(0.013)	(0.009)	(0.012)	(0.014)
Preferences/Tastes	0.011**	0.032***	0.024***	0.032***	0.009
	(0.005)	(0.011)	(0.007)	(0.010)	(0.010)
Background Characteristics	-0.002*	0.0003	0.001	0.0007	-0.0009
	(0.001)	(0.002)	(0.002)	(0.002)	(0.002)

Note: Robust standard errors clustered by school in parentheses. Statistical significance is at: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Panel B reports a Gelbach (2016) decomposition of the difference in the Male coefficient between the baseline and full specification, allocating the total change across pre-defined covariate groups.

Appendix Table 18 reports the subject-level results for practical STEM participation at the Junior Cycle stage. Large gender disparities are evident across all subjects in the baseline specification. The largest gap appears in Material Technology, where the male coefficient is 36.02 percentage points, followed by Technical Graphics (27.33) and Metalwork (16.29). Technology exhibits a smaller but still notable gender gap of 9.00 percentage points.

Introducing the full set of age-9 covariates leads to only modest changes in the estimated gender coefficients. For Technology, the coefficient falls slightly from 9.00 to 8.49 percentage points. Material Technology and Metalwork show marginally larger reductions, from 36.02 to 32.96 and from 16.29 to 14.56, respectively, though substantial gender gaps remain. In Technical Graphics, the estimated male coefficient increases slightly in magnitude, from 27.33 to 28.27 percentage points. Overall, including observable characteristics measured before secondary school entry explains only a small fraction of the gender differences in subject participation.

Consistent with the aggregate Junior Cycle results, the explanatory power of the observed variables is limited across subjects. Preference-related variables account for the largest share of the explained component in Material Technology and Metalwork, contributing 3.38 and 1.99 percentage points, respectively, to the reduction in the gender coefficient. However, these effects remain small relative to the overall gender gaps observed in these subjects. The skills/comparative advantage group contributes very little to explaining the observed disparities. Their estimated contributions are small in magnitude and, across subjects, generally statistically insignificant. Background characteristics likewise account for almost none of the change in the gender coefficients.

Once again, the subject-level results reinforce the broader conclusion that the gender gap in practical STEM participation at the Junior Cycle stage cannot be explained by differences observable at age 9. The mechanisms captured in our data account for only a very small share of the substantial gender disparities observed across practical STEM subjects.

Table 18 JC Practical STEM — Gender gap and Gelbach decomposition - By Individual Subject

	Technology	Material Technology	Metalwork	Tech Graph
Panel A: OLS estimates				
Baseline - Male (1)	0.090***	0.360***	0.163***	0.273***
	(0.017)	(0.026)	(0.017)	(0.022)
Full Model - Male (2)	0.085***	0.330***	0.146***	0.283***
	(0.016)	(0.025)	(0.017)	(0.025)
Observations	3,888	3,888	3,888	3,888
Panel B: Gelbach decomposition of the change in the gender gap				
Total Change: (1)–(2)	0.005	0.031***	0.017***	-0.009
	(0.006)	(0.004)	(0.008)	(0.010)
Contribution to change:				
Skills/Comparative Advantage	0.0007	-0.005	-0.005**	0.003
	(0.002)	(0.010)	(0.002)	(0.002)
Preferences/Tastes	0.007	0.034***	0.020***	-0.010
	(0.006)	(0.009)	(0.008)	(0.009)
Background Characteristics	-0.003	0.002	0.003	-0.003
	(0.002)	(0.004)	(0.002)	(0.003)

Note: Robust standard errors clustered by school in parentheses. Statistical significance is at: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Panel B reports a Gelbach (2016) decomposition of the difference in the Male coefficient between the baseline and full specification, allocating the total change across pre-defined covariate groups.

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